LECTURE NOTES ON AGT 126: MICRO LIVESTOCK PRODUCTION

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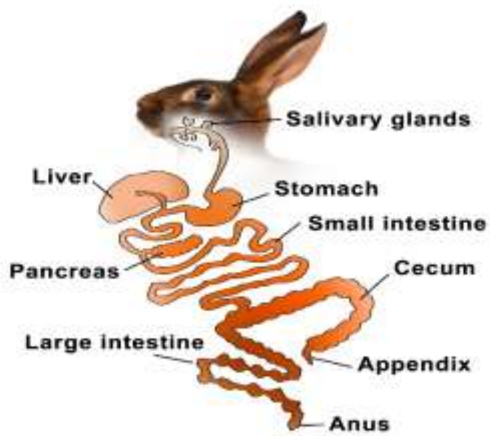
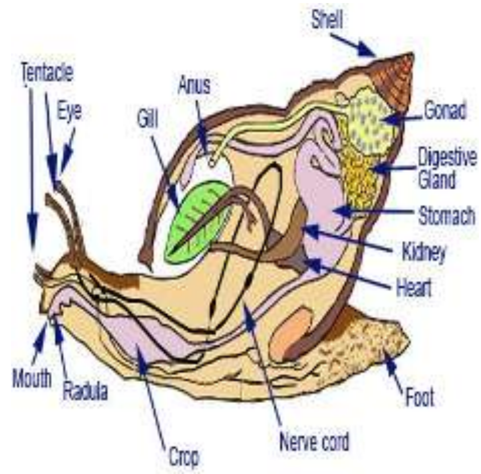
Some Micro Livestock Digestive System

DIGESTIVE SYSTEM OF A RABBIT



DIGESTIVE SYSTEM OF A SNAIL



A QUAIL BIRD

A GUINEA FOWL

A TYPICAL RABBIT

A TYPICAL SNAIL

FEEDING RESOURCES USED FOR MICRO LIVESTOCK

1. Green Forages and Browse Plants

These are fresh, leafy plant materials fed directly to micro-livestock.

- **Examples:**
  - *Rabbits/Guinea pigs/Grasscutters:*
    - Elephant grass (*Pennisetum purpureum*)
    - Guinea grass (*Panicum maximum*)
    - Sweet potato leaves (*Ipomoea batatas*)
    - Cassava leaves (detoxified)
    - Tridax (*Tridax procumbens*)
    - Centrosema, Stylosanthes, and other legumes
- **Benefits:** Rich in fiber, vitamins, and minerals; improves gut health.

2. Agro-Industrial By-products

These are leftovers from food processing industries that can be repurposed as animal feed.

- **Examples:**
  - **Cassava peels**
  - **Wheat offal/bran**
  - **Maize bran**
  - **Soybean meal**
  - **Palm kernel cake**
  - **Brewers' dried grains (BDG)**
- **Common Users:** Rabbits, quails, grasscutters, guinea pigs

3. Grains and Energy-Rich Concentrates

Used to provide energy for growth and production.

- **Examples:**
  - Maize (cracked or ground)
  - Sorghum
  - Millet
  - Rice bran
  - Groundnut cake (protein-rich too)

- **Common Users:** Quails, pigeons, rabbits (in formulated feeds)

4. Protein-Rich Feedstuffs

Essential for muscle development, reproduction, and immunity.

- **Examples:**
    - Fishmeal
    - Soybean meal
    - Blood meal
    - Termites and maggots (especially for birds like quails and pigeons)
    - Earthworms (used in snail and rabbit production)
- **Common Users:** Quails, pigeons, rabbits, grasscutters

5. Fruits and Vegetables

Serve as supplementary feeds, especially for hydration and vitamins.

- **Examples:**
    - Watermelon
    - Pawpaw (papaya)
    - Carrot
    - Pumpkin
    - Cabbage
    - Cucumber
    - Banana peels
    - Plantain peels
- **Common Users:** Rabbits, guinea pigs, grasscutters, snails

6. Kitchen and Farm Wastes (Non-toxic)

Low-cost option, especially in backyard production.

- **Examples:**
    - Yam peels
    - Plantain and banana peels
    - Vegetable scraps
    - Leftover rice or maize (cooked)
- **Note:** Should be free from salt, oil, or harmful residues.

7. Animal Protein Sources

Especially important for omnivorous or carnivorous micro-livestock.

- **Examples:**
    - Maggots (black soldier fly larvae)
    - Termites
    - Earthworms
    - Crushed snails/shellfish for calcium
- **Common Users:** Quails, pigeons, grasscutters (occasionally)

8. Formulated Pelleted Feeds

These are commercially or home-formulated rations containing a balance of protein, energy, fiber, vitamins, and minerals.

- **Species-Specific Pellets:**
    - Rabbit pellets
    - Quail grower and layer mash
    - Grasscutter formulated feeds
    - Pigeon mix (grains and seeds)
- **Benefits:** Balanced nutrition, ease of handling and storage.

9. Mineral and Vitamin Supplements

Needed for bone health, metabolism, reproduction, and disease resistance.

- **Examples:**
    - Salt licks
    - Bone meal
    - Oyster shell (calcium source)
    - Commercial mineral-vitamin premixes
- **Common Users:** All species, especially reproductive and growing stock

10. Water

Although not a feed per se, **clean, fresh water** is a vital nutrient for all micro-livestock. It aids digestion, thermoregulation, and waste excretion.

Summary Table of Feeding Resources per Micro-Livestock Type

Feed Type	Rabbits	Quails	Grasscutters	Snails	Guinea Pigs	Pigeons
Green forages	✓	⊘	✓	✓	✓	⊘
Grains/Concentrates	✓	✓	✓	⊘	⊘	✓
Agro-industrial byproducts	✓	✓	✓	⊘	✓	✓
Fruits & vegetables	✓	⊘	✓	✓	✓	⊘
Animal protein (insects)	⊘	✓	✓ (limited)	✓	⊘	✓
Pelleted/formulated feeds	✓	✓	✓	⊘	✓	✓
Minerals and vitamins	✓	✓	✓	✓	✓	✓
Water	✓	✓	✓	✓	✓	✓

Nutrient Requirements and Nutritional Disorders in Micro-Livestock

1. Nutrient Requirements of Various Micro-Livestock Species

Micro-livestock require the six essential classes of nutrients for optimal performance:

Essential Nutrients:

- **Carbohydrates:** Provide energy
- **Proteins:** Build body tissues and support growth
- **Fats:** Energy reserve and fat-soluble vitamin absorption
- **Vitamins:** Regulate body processes (e.g., Vitamin A, C, D, E)
- **Minerals:** For bone development, egg shell, blood, etc. (e.g., calcium, phosphorus)
- **Water:** Most critical nutrient; supports digestion, circulation, and excretion

Species-Specific Nutrient Needs:

Species	Nutrient Requirements
Rabbits	High fiber (roughage), moderate protein, low fat, calcium, vitamins A & D
Grasscutters	High fiber diet, moderate protein, rich in calcium, phosphorus, and energy

Species	Nutrient Requirements
Snails	High calcium (for shell formation), low protein, fresh greens, and high humidity levels
Quails	High protein (20–24%), energy-rich grains, calcium, vitamin D3 for egg production
Guinea pigs	Fiber-rich feed, **require dietary Vitamin C**, moderate protein

2. Daily Feed and Water Intake per Species

Species	Daily Feed Intake	Daily Water Intake
Rabbits	100–150 g dry feed	200–500 ml
Grasscutters	500–800 g fresh forage	400–600 ml
Snails	20–30 g fresh vegetables	2–5 ml (or from environment)
Quails	20–30 g per bird	50–100 ml
Guinea pigs	80–100 g fresh feed or pellets	100–250 ml

Water needs increase during hot weather, reproduction, and lactation.

3. Symptoms of Nutritional Diseases and Disorders in Micro-Livestock

When any of the essential nutrients are lacking, micro-livestock can suffer from nutritional diseases or metabolic disorders. These impact growth, reproduction, and survival.

Common Nutritional Disorders and Their Symptoms

Deficiency	Symptoms	Species Affected
Protein Deficiency	Poor growth, muscle wasting, rough coat/fur, low reproduction	Rabbits, grasscutters, quails
Vitamin A Deficiency	Eye problems, weak immune response, dry skin/fur	Rabbits, quails
Vitamin C Deficiency	Swollen joints, bleeding gums, poor wound healing, weakness	Guinea pigs (cannot synthesize Vit C)
Calcium Deficiency	Soft/brittle bones, weak shells (snails/quails), poor laying	Snails, quails, rabbits

Deficiency	Symptoms	Species Affected
Phosphorus Deficiency	Weak muscles, poor growth, stiff joints	Grasscutters, guinea pigs
Energy Deficiency	General weakness, stunted growth, poor fertility	All species
Overfeeding (Obesity)	Inactivity, fatty liver, low fertility, joint strain	Rabbits, guinea pigs
Dehydration	Dry mouth, sunken eyes, low feed intake, slow movement	All species

4. Prevention and Control of Nutritional Disorders

- Provide **balanced diets** using species-appropriate feed formulations.
- Always ensure **clean water** is available.
- Supplement with **vitamin-mineral premixes**, especially during stress or disease.
- Feed **fresh forages** rich in fiber and minerals.
- Monitor animal behavior and physical appearance regularly for early signs of deficiency.

Understanding the Reproductivity Micro-Livestock Production

Reproduction is the biological process by which animals produce offspring to ensure the survival and continuity of their species. Good reproductive performance ensures increased production of meat, eggs, and young animals for replacement or sale.

Key Reproductive Terms:

- **Sexual maturity** – Age at which the animal is capable of reproducing
- **Gestation period** – Time between mating and birth
- **Incubation period** – Time it takes for fertilized eggs to hatch (in birds)
- **Litter size** – Number of offspring produced at birth
- **Mating system** – The pattern of pairing between males and females

2. Reproductive Characteristics of Common Micro-Livestock

Species	Sexual Maturity	Gestation/Incubation Period	Litter/Clutch Size	Reproductive Features
Rabbits	4–5 months	28–32 days	4–12 kits	Frequent breeders; induced ovulation
Grasscutters	6–9 months	140–155 days	2–6 pups	Polygamous; seasonal breeders
Snails	6–12 months	21–30 days (egg hatching)	50–200 eggs	Hermaphroditic (both sexes in one snail)
Guinea pigs	3–4 months	59–72 days	2–6 pups	Year-round breeders
Quails	6–8 weeks	16–18 days (egg incubation)	6–12 eggs per clutch	High egg producers; need light stimulation

3. Mating Procedures in Micro-Livestock

Each species of micro-livestock has a unique mating process that must be carefully managed for successful breeding.

A. Rabbits

- **Mating System**: Controlled (1 buck to 5–10 does)
- **Procedure**:
    - Always take the **doe to the buck's cage**.
    - Allow the buck to mate; fall-off indicates successful mating.
    - Remove the doe after mating to avoid aggression.
- **Note**: Rabbits can breed again 1–2 weeks after kindling (giving birth).

B. Grasscutters (Cane Rats)

- **Mating System**: Colony system (1 male to 4–6 females)
- **Procedure**:
    - Form breeding colonies with mature animals.
    - Allow natural mating within the group.
    - Remove unproductive or aggressive males.
- **Note**: Pregnancy signs include swollen abdomen and weight gain.

C. Snails

- **Mating System**: Hermaphroditic (each snail has male and female organs)
- **Procedure**:
  - Keep mature snails (6 months and above) in a moist, dark environment.
  - Snails exchange sperm and both partners can lay eggs.
  - Eggs are laid in soft soil and hatch in 3–4 weeks.
- **Note**: Snails reproduce better in the rainy season.

D. Guinea Pigs

- **Mating System**: Polygamous (1 male to 2–3 females)
- **Procedure**:
  - Introduce the male into the female enclosure.
  - Natural mating occurs quietly and mostly at night.
  - Pregnant females should be separated to avoid stress.
- **Note**: Females should not have their first pregnancy after 8 months of age to avoid birthing problems.

E. Quails

- **Mating System**: Polygamous (1 male to 3–5 females)
- **Procedure**:
  - Maintain mating groups in breeder cages.
  - Natural mating occurs as males mount the females.
  - Fertilized eggs are collected daily and incubated artificially or naturally.
- **Note**: Fertility is influenced by light (14–16 hours/day improves egg production).

4. Importance of Proper Mating Management

- Ensures regular and high-quality offspring production
- Prevents inbreeding and infertility
- Helps in selecting genetically superior breeding stock
- Enhances productivity and farm profitability

Conclusion

A sound understanding of the reproductive behavior and mating procedures of micro-livestock is key to successful production. Each species has specific mating strategies, and careful management ensures high productivity, good health, and sustainable operations.

Quick Revision Questions

1. At what age do rabbits reach sexual maturity?
2. Why must the doe be taken to the buck's cage during mating?
3. What type of reproductive system do snails have?
4. State the average gestation period of grasscutters, rabbits and quail.
5. Why should guinea pig females not breed for the first time after 8 months of age?